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A direct test of the homevoter hypothesis

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Abstract

We propose a methodology that facilitates a direct test of the homevoter hypothesis, which posits that homeowners vote in favor of public projects they perceive increase residential property values and against those that do not. First, we estimate how pre-referendum events that signal a higher probability that the public project will be undertaken impact local residential property values before the referendum is held. These pre-referendum impacts are considered noisy signals to homeowners about the market's assessment of the net marginal benefits of the project. Second, we aggregate these market signals to the precinct level and relate them to precinct-level voting results concerning the proposed project. We apply this two-step approach to the 2004 referendum in Arlington, Texas, for a publicly subsidized stadium for the NFL Dallas Cowboys. The analysis supports the homevoter hypothesis and establishes a possible methodology for future evaluations in this small but growing empirical literature.
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1. Introduction and motivation

Standard voting models assume voters show more support for public spending projects when the expected marginal consumption benefits exceed the marginal costs. When the net benefit of the project influences the value of voters' assets, this wealth effect is expected to influence voting behavior. The capitalization of local public goods into residential property prices has been well established in the economics literature[1]; both the models of Wildasin (1979) and Sonstelie and Portney (1980) show that voters prefer public goods offered at levels that maximize the value of their house, *ceteris paribus*. If the amount of a public good is not offered optimally from a purely consumption perspective, the voter can sell the house and relocate to another jurisdiction in which public goods more closely match the preferences of the voter.[2]

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[1] Oates (1969) is the first empirical work on the capitalization of the level of public spending.

[2] Brueckner and Joo (1991) show that with imperfect mobility, consumption effects will enter into the voter's calculus, although this is less so the earlier the voter expects to leave the community.

Voter support for local public goods or services that preserve or enhance residential property values has been coined the "homevoter hypothesis" by Fischel (2001). Fischel considers homeowners as shareholders in municipal corporations. Unlike their stockholder counterparts, homeowners cannot easily diversify their assets; their home likely constitutes the majority of their wealth. Moreover, homeowners find it harder to move, relative to non-home owners, because of higher transaction costs. These characteristics make it more likely that voting and political activism is more important to homeowners. Specifically, according to the homevoter hypothesis, homeowners are more likely to vote for (against) proposed public goods perceived to increase (decrease) residential property values.

To date, there have been only three empirical tests of the homevoter hypothesis. Brunner et al. (2001) examine voter behavior in California's 1993 school voucher initiative. The initiative would have subsidized private elementary and secondary schools, and hence would have decreased the willingness to pay for housing in quality public school districts. They estimate the premium or discount associated with each of the 74 school districts in Los Angeles County. They establish a negative correlation between the premium paid for housing and support for the school choice initiative, which suggests that homeowners who thought their property values would be harmed by the school choice initiative voted against the proposal. In a follow-up paper, Brunner and Sonstelie (2003) use survey data from potential voters regarding California's 2000 voucher initiative. Their finding that homeowners without school children but in good public school districts were less likely to vote for the initiative than if they lived in inferior school districts lends further support to the homevoter hypothesis.

In a different approach, Hilber and Mayer (2006) test the homevoter hypothesis by examining the relationship between school district spending and the elasticity of supply of vacant residential land. Using data from 46 states, they find that districts having lower percentages of vacant to developed residential land tend to spend more on public education. They argue that quality of education is capitalized in house prices and that in areas where there is less potential for future development, education quality is capitalized in house prices to a greater extent than in areas where there is more potential for new residential development. Their evidence suggests that homeowners consider this tradeoff and are more prone to vote for increases in education spending when they stand to gain more through an increase in the value of their property, *ceteris paribus*.

This paper adds to this small empirical literature by providing the first direct empirical test of the homevoter hypothesis in the context of a large discrete project. We examine voting in a referendum for a new publicly subsidized stadium in Arlington, Texas, for the Dallas Cowboys National Football League (NFL) team. The public subsidization of sports stadiums is a controversial issue, and the debate over whether a subsidy is justified is often finalized at the ballot box. Stadium proponents highlight the quality of life improvements and economic activity generated by new sports venues, thereby justifying subsidies on the basis of public benefits. Carlino and Coulson (2004, 2006) contend that a stadium, or more specifically a franchise that plays in the stadium, provides non-excludable public benefits such as civic pride and enjoyment from being a fan, for which some residents may be willing to pay a premium. Tu (2005) discusses job creation, increased local spending, and economic revitalization of depressed areas.

Those critical of stadium subsidies counter that public benefits of new stadiums tend to be overstated *ex ante*, and that stadium subsidies primarily provide wealth transfers from the taxpaying public to team owners, players, and those fans who will attend events in the new venue. The debate concerning a publicly funded stadium is often contentious and referenda tend to be relatively narrowly decided. As pointed out by Coates et al. (2006), referenda on stadium and arena subsidies have met with mixed results, suggesting that at least in some cases the majority of voters perceive the costs of publicly subsidized facilities to exceed the benefits.[3] Referenda that are narrowly approved or rejected might reflect greater uncertainty about the actual benefits and costs of the stadium.

On November 3, 2004, the citizens of Arlington, Texas, voted on a proposal to increase local sales and user taxes to contribute $325 million to the construction of a new, retractable roof stadium for the NFL's Dallas Cowboys.[4] The proposal was announced in early August of 2004 and, following the model of other successful referendum campaigns, the subsequent three month campaign for the Cowboys stadium focused on the benefits of hosting the Cowboys and the expected positive impact of a new stadium on the future development of

[3] For example, Major League Baseball's San Francisco Giants were denied a publicly built stadium by several Bay Area cities during the 1990s; the Giants eventually built a majority privately financed stadium on the waterfront in downtown San Francisco.

[4] In December 2006, approximately one year into construction, it was announced that the stadium will cost at last $1 billion. The city's contribution is capped at $325 million.

the city. The stadium proposal was approved by the voters of Arlington, passing by a margin of 55% to 45%.[5]

Our empirical strategy entails a two-step process. First, hedonic analysis of single-family residential property prices within the City of Arlington is employed to estimate the impact of various announcements concerning the certainty that a stadium would be built in Arlington. The hedonic model allows for the impact of the stadium proposal announcements to vary across time and distance from the proposed stadium site. The second step follows the applied public choice literature by relating precinct-level vote results to demographics and the estimated house price effects from two pre-vote announcements, each of which increased the probability that a publicly subsidized stadium would be built in Arlington.

The research design allows us to test how the expected net benefits of a public good project, as reflected in house prices, influence voter behavior. We do so by linking house price changes to an increasing probability that a public good project will be undertaken, while recognizing that the net benefits of the public good project can vary by how far an individual's house is from the proposed project site. According to the homevoter hypothesis, homeowners who expect an increase in their property values are more likely to vote for the proposal. Our empirical findings reveal that the direction and magnitude of house price effects explain voter behavior in a manner consistent with the homevoter hypothesis.

Our approach expands on Brunner et al. (2001) in several important dimensions. First, capitalized house price changes from an uncertain future event proxy for the wealth effect homeowners use in determining whether to support a proposed project. Explaining voting patterns using this wealth effect is a direct test of the homevoter hypothesis. Second, to support the assertion that house price effects matter, we examine whether voter turnout is similarly explained by this wealth effect. Finally, because both wealth effects and consumption effects vary with distance from the proposed stadium site, the spatial element plays prominently in the analysis.

[5] The city of Arlington, Texas, hosts the Texas Rangers Baseball Club, for which the city built a stadium in 1994. To the extent that the existing stadium provides local experience with the expected benefits and costs of a professional sports venue, the margin by which the referendum ultimately passed might have reflected reflect less uncertainty on the part of Arlington voters.

2. The Dallas Cowboys stadium in Arlington: background

In April 2001, the Dallas Cowboys announced they were interested in replacing Texas Stadium, located in Irving, Texas and built in 1972. Discussions concerning several preliminary proposals were tabled after the September 11, 2001 attacks in New York City and Washington DC, and the Cowboys stadium search did not return to public light until late 2003. At this time, the City of Dallas proposed to replace the aging Cotton Bowl with a new retractable roof stadium, paid for with a countywide tax. This proposal was ultimately abandoned in spring 2004.

On July 17, 2004, the mayor of Arlington announced that he had been in negotiations with the team about the potential of building a new stadium in Arlington. On August 17, 2004, the Arlington city council approved a ballot initiative to be decided during the November 3, 2004 general election.[6] The ballot initiative was comprised of two parts. First, that the city would provide up to $325 million in public dollars for land acquisition and construction costs for a new retractable roof football stadium for the Dallas Cowboys. The second allowed the city to increase the local sales tax by one half percent, increase car rental taxes by two percentage points and to increase the hotel occupancy tax by five percentage points; the proceeds from the tax increases would be used to retire the debt incurred for the city's contribution to the stadium's construction. On November 3, 2004, the voters of Arlington approved the ballot initiative. The new stadium is scheduled to open for the 2009 NFL football season.

The Cowboys stadium referendum in 2004 was only one of many that have been held throughout the United States since 1990. The dramatic increase in the number of new stadiums across the four major sports in the United States has been accompanied by a large and well-established literature investigating the impacts of new stadiums on local economies. These include the impact of a new stadium on local development (Campbell, 1999; and Nelson, 2001), local employment and income levels (Baade and Dye, 1990, and Coates and Humphreys, 2003), local tourism and hotel occupancy rates (Lavoie and Rodriguez, 2005), and local tax revenue (Coates, 2006, and Coates and Depken, 2007). In a different vein, several papers have investigated the impact of a new stadium on attendance (for example, Clapp and Hakes, 2005), team winning percentage

[6] We label these three dates "announcement" dates, each indicating increased likelihood of adoption of the stadium proposal (see Table 1).

Table 1
Major announcements concerning Dallas Cowboys stadium site search

	Date	Description
Announcement 1*	July 17, 2004	Arlington's mayor announces he has been in secret negotiations with the team about building a new publicly subsidized stadium near the existing baseball stadium in Arlington.
Announcement 2*	August 17, 2004	Arlington's city council approves a ballot initiative for the November 2004 general election. The ballot initiative asks voters to approve up to $325 million toward land acquisition and construction costs for a new stadium located near the existing baseball stadium in Arlington. The ballot initiative also includes a one half cent sales tax in Arlington as well as additional hotel and car rental taxes.
Announcement 3	November 3, 2004	Arlington voters approve ballot initiative on November 3, 2004, and the additional taxes are instituted on April 1, 2005.

* Pertinent to this study in so much as these announcements increased the likelihood of a stadium being built in Arlington and might have influenced property values before the referendum.

(Quinn et al., 2003), and the financial status of the franchise that plays in the stadium (Depken, 2006). The empirical results consistently show that the impact of a new stadium on local economies is dramatically less than advertised before the stadium is constructed and in some instances might actually be negative.[7]

The literature investigating stadium referenda themselves is relatively sparse. Agostini et al. (1997) were the first to estimate a vote-share model in the context of stadium referenda, focusing on 1989 and 1996 votes concerning public subsidization of a new stadium for the San Francisco Giants baseball team. In both votes, they find several demographic variables to be correlated with greater support for the stadium, including income, education, white-collar employment, and Asian heritage. Moreover, they find that the percentage of support for the stadium proposal increased by approximately 15% when the public subsidy was dramatically reduced in the 1996 proposal, which secured majority support. In an inter-city analysis, Depken (2000) investigates how fan loyalty in professional baseball influences stadium referendum outcomes in host cities. He finds that teams with relatively stronger fan bases, i.e., greater fan loyalty, have a higher probability of securing public financing for a new stadium through the referendum process, but his analysis does not include many of the demographics that are common to vote-share models. Coates and Humphreys (2006) are closest in spirit to the study undertaken here and were the first to empirically investigate how proximity to a proposed stadium influences support for a stadium proposal. They investigate several votes in Green Bay, Wisconsin, and Houston, Texas, concerning renovating existing or building new stadiums. Their study suggests that proximity to the proposed stadium had a significant and positive impact on the relative support for a stadium proposal.

While the existing literature focusing on the outcomes of stadium votes suggests that many elements contribute to the probability of success, one influence that has not been included is the anticipated impact of the new stadium on local residential property values. In the context of the Cowboys stadium search, we test the homevoter hypothesis by estimating the impact of the proposed stadium and subsidy on property values in Arlington leading up to the stadium referendum. We identify three specific announcements concerning the probability that a new stadium would be built in Arlington; Table 1 describes these announcements in detail. The first is the Mayor's announcement of negotiations with the Dallas Cowboys concerning a new stadium; the second is the Arlington City Council approving a ballot initiative for the 2004 general election; and the third is the passage of the stadium referendum. We calculate the estimated dollar impact of the two pre-vote announcements on houses that sold in Arlington during the summer of 2004, and average these estimated dollar effects by voting precinct. We then relate the percentage of votes cast in support of the referendum, and voter turnout, to precinct level demographics, the distance of the precinct's voting location relative to the proposed stadium site, and the estimated price effects.

3. Public good announcements and housing prices: empirical model and results

The stadium proposal has the potential to either raise or lower the attractiveness of residing in a neighborhood. Within a given neighborhood, if residents anticipate a positive amenity effect from the stadium exceeding their additional tax burden, this would cause an

[7] See Siegfried and Zimbalist (2000) for a review of the earlier literature concerning the impact of sports on local economies.

increase the demand for houses in that neighborhood. With a fixed supply of houses, prices would rise and homeowners would experience a positive wealth effect. *Ex ante*, we would expect homevoters to offer more support for the proposal. Similarly, if renters expected to enjoy improved local amenities exceeding their additional sales tax burden, demand for rental properties would also rise. However, because landlords would be able to capture most of this "wealth" effect through higher rental rates, we expect renters' support to be weaker than homeowners'.

Before the vote for the Cowboys stadium referendum, both the revelation that the Mayor of Arlington had been in negotiations with the Dallas Cowboys and the Arlington City Council approving a ballot initiative during 2004 general election increased the probability that a new stadium would be built in Arlington.[8] Following these announcements, we assume that homeowners observe signals from residential property transactions in their immediate neighborhood from which they infer the anticipated net effect of the proposed stadium on the market value of their house. If the housing market responds favorably, everything else equal, the proposed public good can be viewed as contributing a net benefit to the local population. While homeowners do not receive direct signals about the value of their own home unless they put it on the market, we assume they do observe transaction prices of properties in their immediate neighborhood. From these transaction prices, homeowners extract a (noisy) signal about whether the market as a whole expects the proposed project to convey net benefits in their locality.

We anticipate both positive and negative house price effects from the stadium. Positive house price effects from stadium access, any (expected) development around the stadium, and potential revenues from parking and other concessions may extend a few miles from the stadium site. On the other hand, negative house price effects from the stadium could be associated with decreased view, congestion, noise, or crime. The hedonic model of stadium-related amenity effects features a distance function that allows the net effect of the stadium to differ by distance and time while maintaining a continuous house price surface. Since our dependent variable is the logarithm of price, we implicitly assume that the amenity effect is proportional to the house value. The empirical model is specified as:

$$\begin{aligned}\mathrm{Ln}(\mathit{PRICE}_i) &= \Gamma\ \mathit{CHAR}\\ &+ \left[\begin{array}{l}(\beta_0 + \sum_{j=1}^{3}\beta_j)(\mathit{Distmax} - \mathit{StadDist})\mathit{Annc}_j\\ +(\beta_4 + \sum_{j=1}^{3}\beta_{j+4})(3 - \mathit{StadDist})D3\mathit{Annc}_j\\ +(\beta_8 + \sum_{j=1}^{3}\beta_{j+8})(2 - \mathit{StadDist})D2\mathit{Annc}_j\\ +(\beta_{12} + \sum_{j=1}^{3}\beta_{j+12})(1 - \mathit{StadDist})D1\mathit{Annc}_j\end{array}\right]\\ &+ v_i \qquad (1)\end{aligned}$$

which distinguishes between the effects of a vector of housing and neighborhood characteristics, *CHAR*, and the spatial-announcement effects, those involving *StadDist* and $Annc_j$.

We first discuss how our specification treats distance and then discuss how we incorporate the announcements. The distance specification uses $Distmax - StadDist$, where *StadDist* is the distance, in miles, from the property to the proposed stadium site, and *Distmax* is the maximum distance between the Arlington city limit and the proposed stadium site. Therefore, $Distmax - StadDist$ is the distance from the housing unit to the periphery of a circle having a *Distmax* radius and the stadium as its mid-point. Unlike the traditional distance function, this specification reveals whether the stadium is placed in a value peak or crater on the larger urban price surface. The coefficient β_0 in Eq. (1) indicates the percentage change in price associated with being one mile closer to the stadium before any of the pre-vote announcements. The model features three dummy variables, $D1$, $D2$, and $D3$, which indicate whether the property is within one mile, two miles, or three miles of the proposed stadium site, respectively.[9] These variables allow house prices effects from the stadium to vary over distance. Interacting the three distance dummies with $1 - StadDist$, $2 - StadDist$, and $3 - StadDist$, respectfully, places kinks or "knots" in the distance function at one, two, and three miles from the proposed stadium site. Pre-announcement, the coefficients β_4, β_8, and β_{12} reveal any additional percentage change in house price per mile closer to the stadium within 3, 2, and 1 miles from the stadium, respectively. Together these four coefficients reveal whether the stadium is placed in a local value "crater" or value "peak" on the larger Arlington house price surface.

[8] The first stage of our empirical approach is an extension of Dehring et al. (2007a) in which five specific announcements concerning the broader search for a stadium site for the Dallas Cowboys in 2004 is investigated. Their inter-city analysis utilizes a differences-in-differences identification scheme within a hedonic pricing model to estimate the average house price effect within Arlington relative to surrounding cities. The announcement dates used in that study which pertain to Arlington are those presented in Table 1. As only the city of Arlington is included in the empirical analysis undertaken herein, a difference-in-difference approach is not necessary.

[9] Price effects beyond three miles were tested, however none were significant, consistent with Tu (2005).

The primary variables of interest in the distance specification are those concerning the stadium announcements. Time of sale relative to a particular stadium announcement is indicated by the dummy variables $Annc_j$, $j = 1, \ldots, 3$, which take a value of one if the property sale was negotiated after the associated stadium announcement, assuming 30 days to closing. The coefficients β_1 and β_2 measure the marginal effect of additional proximity after the Mayor's announcement concerning negotiations with the Dallas Cowboys and after approval of the ballot initiative, respectively. The coefficient β_3 reflects the marginal effect of additional proximity to the stadium site after the stadium referendum passed. A positive coefficient on β_1, β_2 or β_3 would indicate gradient of price with distance increased with stadium announcements 1, 2, or 3 for houses beyond 3 miles of the stadium. The remaining coefficients in the distance specification reveal any additional change in the price gradient following announcements for houses within one mile, two mile, or three miles from the proposed stadium site, respectively. By construction, more of these dummy variable interactions "turn on" as one gets nearer to the stadium site and as more announcements occur so that the sum of all 12 distance coefficients reveals the total percentage change in price per mile within a mile of the stadium following the stadium referendum.

The base data for the hedonic price model, which include sale price, date of sale, and housing characteristics, were obtained from the Dallas-Fort Worth Multiple Listing Service. Elementary school TAKS (Texas Assessment of Knowledge Skills) results were obtained from the Texas Education Agency, and distance variables were generated by geo-coding each residential property's address to latitude–longitude coordinates and calculating the distance from the property to various points of interest. The census block group for each address was identified and key neighborhood characteristics were obtained from the 2000 decennial census. The original sample of Arlington properties had 4073 observations from the 2004 calendar year. After the MLS data were merged with parcel data from the Tarrant County Assessor's Office, seven observations were dropped because of no year or a clearly incorrect year was recorded for when the structure was built, and 902 observations were dropped because they had no lot size recorded; the majority of these were likely condominiums. Ultimately, this stage of the analysis employed a sample of 3108 single family detached houses that sold between January 1, 2004 and December 31, 2004. The sample characteristics are reported in Table 2.

To estimate the hedonic pricing model we apply a logarithmic transformation to the sale price model in Eq. (1). Regression results are presented in Table 3. The parameter estimates on property and neighborhood characteristics are largely as expected.[10] As depicted in Fig. 1, the estimation results indicate that the proposed stadium was located in a localized value crater. The stadium had differential effects on average house prices based on how far the property was from the proposed stadium site. Negative externalities appear to dominate with a half-mile of the stadium but turn positive further out.

The estimation results reveal that for every mile closer to the stadium there was an additional 9% decrease in average house price within two miles of the proposed stadium site, and an additional 17% decrease within one mile of the site. Thus, the proposed stadium site was in the center of a value crater on the broader price surface of Arlington.

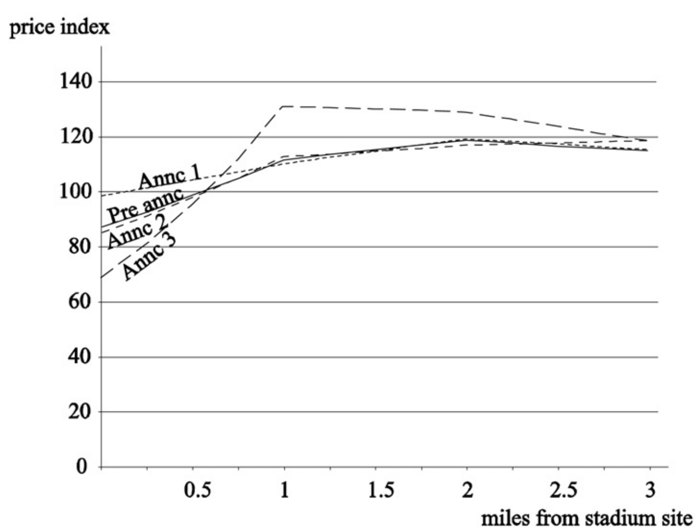
In looking at these intra-city effects, the referendum's passage seems consistent with the homevoter hypothesis. That is, assuming equal density of development around the stadium, the net aggregate change in property value from the stadium announcements is positive. This is seen in Fig. 1 by comparing pre-announcement house prices with those following announcement three. Negative externalities dominate very close to the stadium, while a positive amenity effect dominates farther out. That market prices reflect anticipated net negative amenities in the immediate vicinity of the stadium is not surprising. The cumulative loss in value for properties on the edge of the proposed stadium site was 18.2% over the sample period.[11]

[10] House prices decrease with each additional year of age. Each additional square foot of living space contributes 0.03% to house price. Having a pool increases house price by 10%, while, each additional bath adds 6%. A parking space contributes 3% to price, while each additional story reduces price by 6% (controlling for house size). House prices are 0.66% higher with each additional percent of elementary students rated commendable on the TAKS test in the property's elementary school district. The lot area elasticity of value is 0.16, suggesting that prices increase with acreage but at a decreasing rate. Finally, houses in "white" and "young" neighborhoods have higher prices.

[11] It is important to note that the intra-city analysis does not reveal the total costs or benefits from the stadium announcement. This is because any costs or benefits borne equally by all Arlington residents are not testable in this framework. The intercity analysis of Dehring et al. (2007a) suggests that there was an average reduction in property values in Arlington relative to the surrounding markets that would not bear any tax burden for the new stadium. In the concluding comments we address this puzzle.

Table 2
Descriptive statistics of Arlington house sales

Variable	Description	Mean	Std. Dev.	Min	Max
Price	Sales price ($)	134,113	64,064	23,800	900,000
Square Feet	Square footage	1957	692	670	5800
Baths	Number of bathrooms	2.29	0.71	1	7
Age	House age (years)	21.24	15.15	0	77
Acres	Lot size in acres	0.20	0.14	0.03	3.07
Pool	Pool on property (1 = Yes)	0.14	0.35	0	1
Parking	Number of covered parking spaces	1.79	0.66	0	6
Stories	Number of stories	1.22	0.44	1	3
Owner Occupied at Sale	Owner occupied (1 = Yes)	0.46	0.50	0	1
Vacant at Sale	Vacant (1 = Yes)	0.37	0.48	0	1
PctAcceptable	Percentage of third grade students that scored acceptable on Texas Assessment of Knowledge Skills test	70.71	13.89	23.00	94.00
PctCommendable	Percentage of third grade students that scored commendable on Texas Assessment of Knowledge Skills test	14.95	7.30	1.00	30.00
StadDist	Distance from the proposed stadium site in Arlington measured in miles	4.81	2.57	0.28	10.63
Distmax − StadDist	Maximum distance from the stadium site to the city boundary – actual distance from the stadium site measured in miles	7.91	2.57	2.08	12.44
Distance Fort Worth	Distance from Fort Worth CBD	14.16	2.40	8.43	18.26
Distance Dallas	Distance from Dallas CBD	20.59	2.72	13.87	26.01
PctWhite	Percentage of census block that is white	69.76	15.74	15.18	98.17
PctOver65	Percentage of census block that over 65	5.78	5.61	0.00	30.98
Median Income	Median income of census block ($)	58,111	18,664	17,689	110,219
PctHomeowners	Percentage of census block that owns home	71.39	28.31	0.00	100.00
PctUnemployed	Percentage of census block that is unemployed	4.11	3.33	0.00	20.14
Obs.	3108				



For each period, the price gradient from the stadium site is a continuous, piecewise linear function. Announcement 1 corresponds to the announcement that the city of Arlington was in negotiations with the Dallas Cowboys concerning a new stadium; Announcement 2 corresponds to the Arlington City Council approving a ballot initiative for the general election of November 2004. Announcement 3 corresponds to the date of the November election. Prior to the first announcement house prices were lower nearer to the eventual stadium site. Houses nearest the stadium appreciated with the first announcement but then fell as the proposal became more certain. Houses between one and three miles from the stadium site eventually saw their prices appreciate.

Fig. 1. House price effects from stadium announcements by distance.

Table 3
Housing price regression results

Variable	Parameter estimate	Variable	Parameter estimate
Age	-0.011^{***}	Distmax − StadDist	0.015
	(0.001)		(0.013)
Age Squared	6.5e–05***	(Distmax − StadDist) × Annc 1	0.001
	(0.000)		(0.002)
Square Feet	3.192–04***	(Distmax − StadDist) × Annc 2	0.003
	(0.000)		(0.003)
ln (Acres)	0.029^{***}	(Distmax − StadDist) × Annc 3	−3.06e–04
	(0.005)		(0.004)
Pool	0.092^{***}	1 − StadDist	-0.189^{***}
	(0.009)		(0.073)
Baths	0.057^{***}	(1 − StadDist) × D1 × Annc 1	0.156
	(0.008)		(0.194)
Parking Spaces	0.025^{***}	(1 − StadDist) × D1 × Annc 2	−0.210
	(0.005)		(0.212)
Stories	-0.064^{***}	(1 − StadDist) × D1 × Annc 3	-0.414^{*}
	(0.009)		(0.224)
Owner Occupied at Sale	0.050^{***}	2 − StadDist	-0.095^{**}
	(0.008)		(0.047)
Vacant at Sale	-0.038^{***}	(2 − StadDist) × D2 × Annc 1	−0.016
	(0.008)		(0.099)
PctAcceptable	-0.001^{**}	(2 − StadDist) × D2 × Annc 2	0.090
	(0.001)		(0.109)
PctCommendable	0.005^{***}	(2 − StadDist) × D2 × Annc 3	−0.049
	(0.001)		(0.116)
Distance Fort Worth	−0.017	3 − StadDist	0.019
	(0.013)		(0.031)
Distance Dallas	−0.009	(3 − StadDist) × D3 × Annc 1	−0.004
	(0.016)		(0.043)
PctWhite	0.125^{***}	(3 − StadDist) × D3 × Annc 2	−0.050
	(0.036)		(0.047)
PctOver65	-0.265^{***}	(3 − StadDist) × D3 × Annc 3	0.099^{*}
	(0.088)		(0.051)
Median Income	1.47e–07	Constant	11.119
	(0.000)		(0.615)
PctHomeowners	−0.004	Observations	3108
	(0.025)	R-squared	0.85
PctUnemployed	0.167	F-statistic	221.33^{***}
	(0.110)		

Notes. Dependent variable is the natural logarithm of house price. Annc 1 corresponds to the announcement that the city of Arlington was in negotiations with the Dallas Cowboys; Annc 2 corresponds to the Arlington City Council approving a ballot initiative for the general election in November 2004; Annc 3 corresponds to the date of the November election. D1, D2, and D3 are dummy variables that take a value of one if the property is within one mile, two miles, or three miles of the proposed stadium site, respectively, and zero otherwise. Month dummy variables included but not reported. A GLS estimator is employed that allows for heteroskedasticity by zipcode. Variables defined in Tables 1 and 2. Standard errors in parentheses.

* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

4. Voting patterns and public project announcements

In the second phase of our empirical analysis, we investigate whether the perceived costs and benefits of the stadium, as capitalized into house prices during the pre-vote period, affected voting behavior with regard to the Cowboys stadium referendum. Specifically, we relate precinct-level support and voter turnout to demographic characteristics of the precinct and the average estimated effect of the two pre-vote announcements on house prices in the precinct. Of particular interest is whether the average effects on housing prices in a precinct help explain voting patterns and voter turnout.

The homevoter hypothesis predicts a direct relationship between price effects attributable to the proposed public good project and support for the project at the voting booth, i.e., homevoters will offer more support for public good projects that provide a greater net increase the value of their property. We estimate the dollar impact of Announcement 1 and Announcement 2, net of the house characteristics and neighborhood effects, on the sale price of each transacted property in our sample using the results from the hedonic regression model presented above.

Specifically, we use the hedonic model estimates to calculate the impact of proximity to the proposed stadium site during each pre-vote announcement period. For example, if a property located within one mile of the stadium sold after the Mayor's announcement but before the City Council's approved the ballot initiative, i.e., between Announcement 1 and Announcement 2, the *additional* percentage change in price per mile of distance closer to the stadium site is calculated as $\beta_1 + \beta_5 + \beta_9 + \beta_{13}$ from Eq. (1). For each property that sold between Announcement 1 and Announcement 2, and between Announcement 2 and Announcement 3, we calculate the impact of the announcements on sales price; if a property sold before Announcement 1 or after Announcement 3, it is not included in these calculations.

Each property was then geo-coded to the Arlington voting precinct in which it was located, and then for each pre-vote announcement period and precinct the average dollar impact is calculated. These average price effects are combined with other precinct-level demographics in a voting model similar in spirit to that used in Brunner et al. (2001):

$$\begin{aligned}
PctYES_i = {} & \theta_0 + \theta_1 PctTransient_i + \theta_2 PctOver65_i \\
& + \theta_3 Income_i + \theta_4 PctUnemployed_i \\
& + \theta_5 PctWhite_i + \theta_6 PctHomeowner_i \\
& + \theta_7 StadDist_i + \theta_8 PriceChg12_i \\
& + \theta_9 ZeroSales12_i + \theta_{10} PriceChg12_i \\
& + \theta_{11} ZeroSales23_i + \omega_i \qquad (2)
\end{aligned}$$

where the dependent variable is the percentage of total votes cast in favor of the stadium proposal in precinct i, and ω is a zero-mean stochastic error term. The variable *PctTransient* is the percentage of a precinct's population that was in another state in 1995. An individual living in another state in 1995 was less likely to have participated in the 1991 referendum in Arlington to build a stadium for the Texas Rangers baseball club. Although the promised economic development around the stadium had not materialized by the time the Cowboys proposal was being considered, the memory of such promises would be strongest for those who were living in Arlington at the time of the baseball stadium referendum. Moreover, a precinct with a greater proportion of the population that lived in another state in 1995 might be comprised of fewer people likely to live in Arlington for the entire life of the proposed stadium, thereby reducing their expected tax contribution to the stadium. For both of these reasons we expect *Transient* to be associated with a greater level of support for the stadium proposal. The expected effect of the percentage of the precinct's population over the age of sixty-five, *PctOver65*, is ambiguous. On one hand, older residents often have fixed incomes and are somewhat reluctant to vote for tax increases, but in this case those over sixty-five years old might have a lower expected contribution to the stadium. The net effect of these influences is not clear.

Depken (2001) shows that professional football is a normal good in cities that host football franchises. If this is also the case in Arlington, those with greater income are expected to attend more football games and, relative to those who do not attend games in the new stadium, would earn more consumer surplus from the new football stadium. Although those with more income might bear a greater tax burden, we expect that, on net, support for the stadium will be higher in precincts with wealthier populations; therefore, we expect a positive parameter estimate on the variable *Income.* Areas with more unemployed individuals are less likely to anticipate large consumer surplus from a new stadium and are simultaneously more reluctant to vote for increases in potentially regressive sales taxes. Therefore we expect a negative parameter estimate on the variable *PctUnemployed*. The impact of racial composition on stadium support is ambiguous.

The primary variables of interest in the vote model are the distance of the precinct from the proposed stadium site, *StadDist*, and the estimated impacts of the various stadium announcements on residential property values. Following Coates and Humphreys (2006), we anticipate that the further the voting precinct is from the proposed stadium site, the lower the support for the stadium, *ceteris paribus*. The homevoter hypothesis predicts that precincts with higher positive (lower negative) house price changes during a given announcement period would exhibit more (less) support for the proposed Cowboys stadium, *ceteris paribus*. The estimated price effects are differentiated as those that occurred between

Announcement 1 and Announcement 2, *PriceChg12*, and those that occurred between Announcement 2 and Announcement 3, *Pricechg23*. If the behavior of Arlington voters was consistent with the homevoter hypothesis, we anticipate a positive coefficient on estimated price effects.

Not all precincts in the city of Arlington had a house sell during a given announcement period. To accommodate this we create two dummy variables that take a value of one if there were no house sales during a given pre-vote announcement period. *ZeroSales12* refers to the period between Announcement 1 and Announcement 2, and *ZeroSales23* refers to the period between Announcement 2 and Announcement 3. If a precinct had no home sales during a given announcement period, the dollar effect associated with that announcement period is coded as zero and the dummy variable differentiates between a price effect that equals zero and the absence of a price effect on voting patterns. If homeowners receive no information about the impact of the stadium proposal on local property values, that is one or both of the *ZeroSales* dummy variables takes a value of one, the impact on support for the proposal is ambiguous.

Combining the results heretofore with the inter-city price changes documented by Dehring et al. (2007a), it is of interest whether precincts with a greater proportion of homeowners provided less support for the proposed Cowboys stadium, *ceteris paribus*.

For renters and homeowners, the choice of whether to vote in the affirmative for the stadium is a combination of three (interrelated) effects: taxation, amenities, and property values. Renters and homeowners respond similarly to taxes and amenities: positive amenities increase support, negative amenities reduce support, and increased taxation would reduce support, ceteris paribus. The influence of changes in property values on an individual's support for the stadium might be opposite for renters and homeowners: reductions in property values reduce support amongst homeowners and might increase support amongst renters, if they anticipate a reduction in rental rates. The net of these three effects influences the choice of whether to support the stadium or not.

As we do not have data on individual renter and homeowner characteristics, and therefore cannot employ the methodology of Brunner et al. (2001), we include the percentage of a precinct's population that are homeowners (*PctHomeowners)* to test whether homeowners, as a group, were less prone to support the stadium proposal. Additionally, the percentage homeowners was interacted with the price effect for each pre-vote announcement period and included in the voting model; the coefficients on the interaction variables represent the degree to which homeowners responded differently to the proposed stadium. However, neither of the interactions proved statistically significant and they were therefore dropped from subsequent models.

The total number of votes cast concerning the stadium proposal and the total votes in the affirmative were obtained for each of 123 voting precincts in the city of Arlington from the Tarrant County election commission.[12] Election totals were available for early voting, which took place from October 1 through October 29, 2004, and voting that took place on the day of the general election, November 3, 2004. In this analysis, we combine early and day-of-election votes to generate the total number of votes from which we determine the percentage of affirmative votes in a precinct voting. We matched 113 of the 123 voting precincts in Arlington with U.S. census tract data from the 2000 census, from which we obtained the population characteristics, income, unemployment, the percent white, and the percentage of residents that were homeowners.[13] Distance is calculated from the proposed stadium site to the polling location of each precinct.

Descriptive statistics of the 113 voting precincts in the city of Arlington employed in our regression model are reported in Table 4. Average support for the stadium proposal throughout all of Arlington was 56.7%. However, there was considerable variation across precincts. Twelve precincts offered less than 50% support for the stadium proposal. Ten percent of the average precinct's population was in another state in 1995, approximately 6.75% of the population was older than sixty-five, median household income was approximately $55,000, the unemployment rate averaged 3%, the average precinct was 64% white, the average precinct was 7.5 miles from the proposed stadium location, and 61% of each precinct's population owned their house.

The sample distribution of sales by precinct and distance from the stadium is reported in Table 5. A precinct is identified spatially based on the location of the polling place. A district may include sales from more than one distance range if the district boundaries are not con-

[12] The precincts not included in our final sample include two that were technically located in Kennendale, a border city with Arlington. The remaining precincts did not report any voting activity during the general election of 2004.

[13] Specifically, we gathered data for the census tract in which the precinct's polling site is located. The data were obtained from the US Census Bureau's (2007) *American FactFinder* (http://factfinder.census.gov), last accessed October 2007.

Table 4
Descriptive Statistics of Arlington Voting Precincts

Variable	Description	Mean	Std. Dev.	Min	Max
PctYES	Percent voting 'YES' on the stadium proposal	56.7	8.9	43.4	100.0
PctTurnout[a]	Percent of registered voters voting on the proposal	62.6	13.0	12.0	83.0
PctTransient	Percent living in a different state in 1995	10.3	5.4	1.1	23.6
PctOver65	Percent over 65	6.7	4.6	1.5	23.9
Income	Median household income (thousand)	54.6	18.4	19.3	104.7
PctUnemployed	Percent unemployed	3.1	1.4	0.6	6.2
PctWhite	Percent white	63.9	18.9	9.0	90.0
PctHomeowners	Percent homeowners	61.2	28.8	4.1	95.7
PriceChg12	Average price effect during period between Announcement 1 and Announcement 2 (thousand $)	0.6	1.0	−2.7	5.8
ZeroSales12	No properties sold between Announcement 1 and Announcement 2	0.3	0.4	0.0	1.0
PriceChg23	Average price effect during period between Announcement 2 and Announcement 3 (thousand $)	2.3	2.2	−0.8	14.3
ZeroSales23	No properties sold between Announcement 2 and Announcement 3	0.2	0.4	0.0	1.0
Observations	113				

Notes. Percentage voting YES includes early and day of election voting returns. Vote data obtained from the Arlington City Clerk's office. Demographic data obtained from US Census.
[a] Based on 93 observations for which the total number of registered voters is available.

Table 5
Distribution of sales within 3 miles of stadium, by announcement period and district

Date of sale	Distance from stadium	Total number of sales	Number of precincts represented	Number of precincts with no sales
Before Announcement 1	2–3 mi	181	16	0
	1–2 mi	299	31	2
	0–1 mi	108	11	0
Between Announcement 1 and Announcement 2	2–3 mi	25	16	5
	1–2 mi	35	31	13
	0–1 mi	14	11	5
Between Announcement 2 and Announcement 3	2–3 mi	67	16	1
	1–2 mi	92	31	5
	0–1 mi	40	11	3
After Announcement 3	2–3 mi	31	16	4
	1–2 mi	25	31	17
	0–1 mi	11	11	7

Notes. A precinct is identified spatially using the location of the precinct's polling place. A district might include sales from more than one distance range if the district's boundaries are not contained within a given distance range. Announcement 1 corresponds to the announcement that the city of Arlington was in negotiations with the Dallas Cowboys concerning a new stadium; Announcement 2 corresponds to the Arlington City Council approving a ballot initiative during the general election of November 2004; Announcement 3 corresponds to the date (outcome) of the November election.

tained within a given distance range. Table 5 shows the number of precincts represented by these sales, as well as the number of precincts in that given distance range having no sales during the announcement period.

The dependent variable in the estimating equation is the proportion of votes cast in the affirmative for the stadium proposal. It is well known that using a proportion as a dependent variable introduces heteroskedasticity, in which case ordinary least squares standard errors are incorrect. We address this problem by allowing for generalized heteroskedasticity using a Huber–White–Sandwich estimator.[14] The primary independent variables of interest are the estimated dollar impacts of the various stadium announcements. However, these variables are generated regressors, which can introduce

[14] An alternative is to use weighted least squares in which the weights are the total votes cast in a particular precinct. We estimated all of our models using weighted least squares with no change in the qualitative results presented here.

Table 6
Support for the Cowboys stadium proposal (dependent variable: percent voting yes)

	(1) PctYES	(2) PctYES	(3) PctYES	(4) PctYES	(5) PctYES	(6) PctYES	(7) PctYES
PctTransient	0.483**	−0.205	0.402**	0.388*	0.414*	0.418**	0.414*
	(0.211)	(0.239)	(0.20)	(0.221)	(0.221)	(0.208)	(0.216)
PctOver65	−0.098	−0.205	−0.056	−0.098	−0.022	0.129	0.119
	(0.223)	(0.286)	(0.30)	(0.320)	(0.238)	(0.297)	(0.322)
Income	0.155**	0.160**	0.134**	0.140**	0.170**	0.153**	0.157**
	(0.064)	(0.070)	(0.07)	(0.071)	(0.069)	(0.075)	(0.077)
PctUnemployed	0.111	0.135	−0.174	−0.112	0.204	−0.041	−0.011
	(0.867)	(0.865)	(0.76)	(0.773)	(0.885)	(0.760)	(0.829)
PctWhite	−0.002	−0.009	−0.061	−0.057	−0.040	−0.093	−0.093
	(0.063)	(0.067)	(0.07)	(0.071)	(0.064)	(0.071)	(0.074)
Distance to Stadium	−0.361*	−0.421*	−0.534**	−0.570**			
	(0.194)	(0.224)	(0.25)	(0.259)			
PctHomeowners	−0.059	−0.022	−0.012	−0.017	−0.066*	−0.068*	−0.075*
	(0.040)	(0.040)	(0.036)	(0.040)	(0.037)	(0.036)	(0.041)
ZeroSales12		6.054**		−1.471	5.822**		−1.416
		(2.97)		(1.998)	(2.820)		(2.045)
PriceChg12		1.073		0.529	0.774		0.233
		(0.83)		(0.555)	(0.760)		(0.553)
ZeroSales23			11.765***	13.261***		11.250***	12.535***
			(3.82)	(4.348)		(3.811)	(4.119)
PriceChg23			1.162***	1.134***		0.954**	0.951**
			(0.43)	(0.420)		(0.421)	(0.411)
Constant	50.046	47.766	50.200	50.258	47.264	49.344	49.566
	(5.565)	(6.175)	(5.286)	(5.724)	(6.161)	(5.645)	(5.931)
Observations	113	113	113	113	113	113	113
R-squared	0.21	0.27	0.38	0.39	0.25	0.35	0.36

Notes. Dependent variable is the percent of total votes cast in favor of the Cowboys stadium referendum. Announcement 1 corresponds to the announcement that the city of Arlington was in negotiations with the Dallas Cowboys concerning a new stadium; Announcement 2 corresponds to the Arlington City Council approving a ballot initiative during the general election of November 2004. Announcement 3 corresponds to the date of the November election. Variables as defined in Table 4. Bootstrapped standard errors in parentheses.

* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

potential bias in standard ordinary least squares standard errors (Murphy and Topel, 1985, and Wooldridge, 2001). We address this problem by bootstrapping the standard errors using 500 replications.

The results of several different specifications are reported in Table 6. Model 1 serves as a benchmark case; it does not include any of the variables concerning property values or property sales. The percentage of the population in another state in 1995 and income are both positively correlated with support for the stadium. Consistent with Coates and Humphreys (2006), support for the stadium declined as the precinct was further from the proposed stadium site. The remaining variables are consistently insignificant.

Model 2 in Table 6 introduces the variables *PriceChg12* and *ZeroSales12*. The parameter estimates on the original set of variables do not change in magnitude or significance. The parameter on *PriceChg12* is positive, but not statistically different from zero at the usual confidence levels, and the parameter on *ZeroSales12* is positive and statistically significant. The lack of significance of *PriceChg12* may not be surprising because any information embodied in these property sales would have been obtained up to six months prior to the actual vote.

Model 3 includes only *PriceChg23* and *ZeroSales23*. The coefficient on *PriceChg23* is positive and statistically significant. Precincts in which property values were increasing during the Announcement 2 period offered more support for the stadium. Specifically, there was a 1.2% increase in support for the stadium for every \$1000 dollar increase in house prices. Moreover, in precincts without any home sales during the Announcement 2 period, support for the stadium was approximately 11 percentage points greater. Model 4 includes *PriceChg12*, *ZeroSales12*, *PriceChg23*, and

ZeroSales23. In this model the variables associated with the first announcement period are insignificant whereas those associated with the second announcement period are significant. The evidence suggests that the impact of the stadium proposal on housing prices was more important (in a statistical and economic sense) during the second announcement period, closer to when the vote took place, than during the first announcement period, months ahead of the vote in the early part of the summer of 2004.

Models 5–7 repeat the estimation dropping the distance variable; there might be concern that the correlation between the distance from the proposed stadium site and the percentage of a precinct that are homeowners is sufficiently high to induce noise in the standard errors of the remaining parameter estimates. In Models 5–7 the results do not qualitatively change, however the percentage of the precinct's population that owned their own home is now negative and statistically significant. The results suggest that property owners did not anticipate the same level of net benefits as non-property owners, perhaps because they feared a greater share of the tax burden or considered themselves less mobile and therefore unable to leave the tax jurisdiction. However, as houses in a particular precinct sold for higher prices between Announcement 2 and Announcement 3, that is, immediately before the stadium referendum, support for the stadium proposal increased.

If voter support for the stadium proposal were explained by the effect of the proposed stadium on house prices, we might also expect the magnitude of the price effect to explain voter turnout. That is, conditional on receiving a signal from the market concerning the expected net benefits of the proposed stadium, the greater the price effect the higher should be voter turnout. Although the stadium proposal was coincident with the presidential election of 2004, a finding that turnout responded to the signals provided by the residential property market would be consistent with our findings that price effects matter. Therefore, we test whether the results reported in Table 6 are confirmed when voter turnout is the dependent variable rather than percentage voting in favor of the stadium proposal.[15]

We obtained information on the number of registered voters for 94 of the 113 precincts in our sample. Knowing the total number of votes cast in each precinct facilitates calculating the percentage of registered voters who participated in the stadium vote. We re-estimated the models in Table 6 replacing percent support with percent turnout. The results are reported in Table 7.

Turnout was statistically higher in precincts with older populations, higher incomes, and where there were more whites. Decreases in house values following Announcement 2 were associated with greater turnout. Models 3 and 4 indicate a 0.98% and 0.93% decrease in voter turnout for every \$1000 dollar increase in house prices. Put differently, districts in which price declines were the largest were more likely to turn out. It is interesting to note that *PctHomeowners* was not a significant determinant of voter turnout. This is in contrast to Dipasquale and Glaeser (1999), although the results may be complicated by the fact that the November 2004 election was a general election. Combined with the results from Table 6, the findings from the vote models are consistent with the homevoter hypothesis; voters consider how proposed public projects are expected to influence their property values in determining their support for the project.

There may be concern that the empirical evidence revealed thus far is spurious. Therefore, we undertook a battery of robustness checks in which we estimated vote models using the outcomes of a number of previous votes in Arlington focusing on proposed public goods and mayoral elections. In the interest of space, the results are not printed here but are available in an extended empirical appendix (Dehring et al., 2007b).

To briefly summarize our robustness checks, we estimated various voting models of the same specification as Eq. (2). We related demographics and the estimated price effects of the stadium proposal announcements to voter support for sales tax increases for mass transit in the form of buses (May 2002 and February 2003) and for street maintenance (November 2003), and in the May 2003 Arlington Mayoral election for Robert Cluck, the mayor who initiated the negotiation with the Dallas Cowboys to relocate to Arlington.

In each case the vast majority of the parameter estimates are insignificant. However, those precincts with a large proportion of older residents tended to vote against the mass transit proposal and in favor of electing Robert Cluck mayor of Arlington more so than in other precincts, ceteris paribus. None of the property-related variables were statistically significant, including the percentage of the precinct's population that was a homeowner.

Overall, the robustness checks suggest that the results presented in Table 6 are not spurious or that the results supporting the homevoter hypothesis are caused by some unmeasured influence on voter behavior. Given the direct test of the homevoter hypothesis undertaken

[15] See Hastings et al. (2007) for a similar analysis focusing on voter participation and education lotteries.

Table 7
Voter turnout as a robustness check (dependent variable: percent turnout)

Coefficient	(1) Turnout	(2) Turnout	(3) Turnout	(4) Turnout
PctTransient	0.024	0.045	0.054	0.062
	(0.350)	(0.320)	(0.290)	(0.280)
PctOver65	0.594***	0.604**	0.447	0.470
	(0.230)	(0.240)	(0.280)	(0.290)
Income	0.401***	0.402***	0.422***	0.418***
	(0.098)	(0.087)	(0.090)	(0.089)
PctUnemployment	−0.172	−0.470	−0.330	−0.392
	(1.420)	(1.380)	(1.080)	(1.180)
PctWhite	0.159	0.146	0.181*	0.177*
	(0.130)	(0.120)	(0.100)	(0.110)
PctHomeowner	0.038	−0.017	−0.005	−0.007
	(0.054)	(0.050)	(0.048)	(0.051)
ZeroSales12		−8.430**		−0.695
		(3.500)		(2.560)
PriceChg12		−0.639		−0.329
		(0.720)		(0.820)
ZeroSales23			−16.813***	−16.410***
			(6.250)	(6.300)
PriceChg23			−0.983**	−0.927**
			(0.430)	(0.470)
Constant	23.086	30.386	28.724	29.451
	(9.520)	(10.700)	(8.380)	(9.220)
Observations	93	93	93	93
R-squared	0.62	0.63	0.70	0.69

Notes. Dependent variable is the percent of eligible voters who participated in the Cowboys stadium referendum. Announcement 1 corresponds to the announcement that the city of Arlington was in negotiations with the Dallas Cowboys concerning a new stadium; Announcement 2 corresponds to the Arlington City Council approving a ballot initiative for the general election of November 2004. Announcement 3 corresponds to the date of the November election. Variables as defined in Table 4. Bootstrapped standard errors in parentheses.

* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

here, we find support for the hypothesis despite the fact that the stadium proposal passed.

5. Discussion and conclusions

This paper contributes to a relatively small but growing empirical literature investigating the homevoter hypothesis. It is the first paper to investigate a large discrete public good project in the context of the homevoter hypothesis. It is also first to provide a direct test of the hypothesis in that voter behavior is tied to *pre-vote* changes in residential property values, which vary both temporally and spatially. Our results indicate that support for a publicly subsidized stadium to host the NFL's Dallas Cowboys in Arlington, Texas was positively related to the estimated impacts of the proposed stadium on local residential property values. Specifically, we find between a 0.9% and 1.2% increase in support for the stadium for every $1000 dollar increase in house prices. We also find that homeowners in general were less likely to support the stadium proposal, and that voters in districts in which there were no house sales were more likely to support the stadium. Finally, we find a 1.0% decrease in voter turnout for every $1000 dollar increase in house prices, indicating that districts in which price declines were the largest were more likely to turn out.

If *average* property values in Arlington fell during the months leading up to the stadium referendum, consistent with Dehring et al. (2007a), the referendum's success would seem in violation of the homevoter hypothesis. One explanation might be a difference between the average homeowner and the median voter. The median voter might vote in the affirmative if other determinants rather than expected changes in property values are considered. For example, individuals who anticipated considerable personal consumption benefits from a new stadium, e.g., because they anticipated attending events in the new stadium, might have been more likely to turn out and to vote 'yes.' Further, those who anticipated considerably smaller costs from building the stadium, e.g., transient or more mobile residents, might have been more likely to vote in the affirmative, everything else equal. Finally, a relatively small net cost may not induce an abstainer to turn out and vote 'no,' and therefore the median voter might have differed considerably from the average homeowner in Arlington.

Another explanation could be imperfect information in voting markets. Public officials may personally benefit from a project even if there are net costs to city residents as a whole and might therefore focus on the perceived or anticipated benefits of the stadium while deemphasizing the costs involved. Indeed, Porter and Thomas (2006) point out that city and team officials often over-promote the anticipated public benefits of a proposed stadium project, including appeals to city pride and notoriety, increased economic activity from tourism, and the potential for hosting mega-events such as a Super Bowl or an NCAA Final Four.

These public benefits have proven difficult to measure and identify; most academic studies suggest the public benefits *after* a stadium is built are considerably less than those predicted *before* a stadium is built. However, to the extent that voters do not completely dismiss these claims as mere 'cheap talk,' such claims might influence the median voter's support for a stadium subsidy. Our results suggest that voters more susceptible to 'cheap talk,' and those with less experience with past similar projects, were more likely to support the stadium proposal.

A related explanation also stems from imperfect information in voting markets. Voters with shorter time horizons and therefore less direct exposure to potential future tax increases are more likely to estimate larger personal benefits and smaller personal costs of a large discrete project such as stadium. These voters may support the proposal even without positive net benefits. Our results are also consistent with this explanation.

While reduced-form in nature, this test of the homevoter hypothesis provides an intuitively appealing explanation for how and why homeowners vote on public-good proposals. When a large, discrete project funded with a considerable outlay of public money is debated, the outcome of a public referendum often hinges upon the public's perception of the costs and benefits of the project. Our empirical results that, on the margin, voters internalized market information concerning the expected net benefits of the stadium conveyed through changing property values associated with an increased likelihood that the stadium would be built.

Consistent with the homevoter hypothesis, regardless of whether property values changed in a voting precinct during the stadium debate, those precincts with a greater proportion of homeowners showed less support for the stadium proposal. Moreover, providing more direct support for the homevoter hypothesis, in precincts where property values fell during the Arlington stadium debate there was an additional reduction in support for the proposal. This suggests that price changes associated with the stadium debate were conveyed in some fashion and altered the expected net benefits of the stadium. Future research in the context of stadium referenda and the homevoter hypothesis would add to our empirical understanding of how homeowners support proposed large, discrete, public good projects.

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